

3.2.P.2 Pharmaceutical Development

AMLOZA (Amlodipine Besylate Tablets, USP) 5 mg / 10 mg — ANDA

3.2.P.2.1 Components of the Drug Product

3.2.P.2.1.1 Drug Substance: Based on published literature and prior use of these excipients in approved amlodipine products, no known incompatibilities have been identified between amlodipine besylate and microcrystalline cellulose, dibasic calcium phosphate, sodium starch glycolate, or magnesium stearate; however, no experimental drug–excipient compatibility data are presented in this academic dossier.

3.2.P.2.1.2 Excipients: Each excipient was selected based on established compendial status, compatibility, and prior use in approved amlodipine besylate products (as referenced in the FDA Inactive Ingredient Database).

3.2.P.2.2 Drug Product

3.2.P.2.2.1 Formulation Development: Literature reports amlodipine besylate as exhibiting high solubility and high permeability, consistent with a BCS Class I profile, with acceptable flow and compression characteristics supporting selection of direct compression over wet granulation; this ANDA is supported by in vivo bioequivalence data (Module 5) rather than a BCS-based biowaiver. Direct compression was preferred to minimize processing steps and reduce exposure of the drug substance to moisture and heat during manufacture, given known sensitivity of the dihydropyridine ring system to hydrolytic and photolytic degradation.

3.2.P.2.2.2 Overages: No overage is proposed; the manufacturing process and analytical controls are designed to ensure label claim is met at time of release and through shelf life without the need for an intentional overage.

3.2.P.2.2.3 Physicochemical and Biological Properties: As a BCS Class I compound, in vitro dissolution is not expected to be the rate-limiting step for absorption; nonetheless, dissolution testing is included in the specification as a standard quality control and batch-to-batch consistency measure, and comparative dissolution profiling against the RLD supports the overall bioequivalence narrative documented in Module 5.

3.2.P.2.3 Manufacturing Process Development

The manufacturing process (direct compression) was selected at formulation development stage and is intended to be maintained from exhibit batch through commercial scale, subject to standard scale-up considerations (blend uniformity at scale, compression tooling change, if any). Any changes identified during scale-up or post-approval would be evaluated and managed in accordance with applicable FDA post-approval change guidance (e.g., SUPAC-IR, 21 CFR 314.70), with the reporting category (Annual Report, Changes Being Effected, or Prior Approval Supplement) determined by the nature and potential impact of the change on product quality.

3.2.P.2.4 Container Closure System

The proposed container closure system (HDPE bottle with child-resistant closure, and/or PVC/aluminium blister) is consistent with RLD presentation and standard practice for immediate-release oral solid dosage forms; suitability is supported by compendial compliance of all packaging materials.

3.2.P.2.5 Microbiological Attributes

As a non-sterile, non-aqueous immediate-release oral tablet, AMLOZA is categorized per USP <1111> as a dosage form for which routine microbial limits testing at every batch release is not typically required, provided appropriate controls are demonstrated during development. Should testing be warranted, total aerobic microbial count and total combined yeasts/molds count would be determined per USP <61>, with absence of specified objectionable organisms confirmed per USP <62>, consistent with the acceptance criteria in USP <1111> for this dosage form category.

3.2.P.2.6 Compatibility

Compatibility studies would typically be conducted during development to confirm the absence of drug-excipient interaction (e.g., via forced degradation / binary mixture studies under accelerated conditions). No experimental compatibility data are presented in this academic dossier.

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